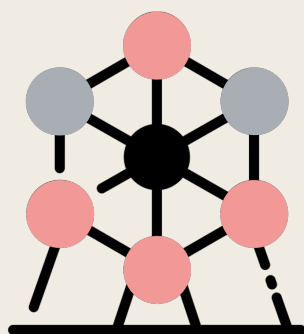




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Programme

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OC02	09:15–09:30	Joan Grèbol-Tomàs	What Does an Atomic Partition Need to be Reliable?
OC03	09:30–09:45	Diogo J. L. Rodrigues	Atoms-in-Molecules Rényi Entanglement Entropies for Correlated Wavefunctions
OC04	09:45–10:00	Ishita Bhattacharjee	Benchmarking TDDFT in Describing Excited State Bonding Analysis
OC05	10:00–10:15	Daniel Barrena-Espés	Exotic Molecules Through the Lens of Quantum Chemical Topology: A Fresh Perspective on Chemical Bonding
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OC07	11:00–11:15	Nil Roig-Vidal	Looking Around the Dewar–Chatt–Duncanson Model: Environmental Effects in TM–CO Bonding
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OC11	12:00–12:15	Marina Descoubes	Computational Elucidation of Ligand Driven Selectivity in Gold(I) Catalyzed Alkyne Hydroamination
OC12	12:15–12:30	Iñigo Iribarren	A Dynamic Evaluation of Cisplatin Encapsulation into $[\text{Pd}_2\text{L}_4]^{4+}$ Metallocages in Solution
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OC14	13:45–14:00	Fiona J. Gibson	Contrasting Effects of Triazole Substituents Used in Hydrogen, Halogen and Chalcogen Bonding
OC15	14:00–14:15	Manjaly J. Ajitha	DFT Insights into Noncovalent Stabilization of Cationic Intermediates in Pillar[5]arene-Catalyzed Anti-Markovnikov Halogenation
OC16	14:15–14:30	Lucas Albouy	Theoretical Study of Monohydrated Molecules Bearing Sulfur and Oxygen Atoms
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Code	Time	Speaker	Title
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OC21	16:00–16:15	Laura N. Schiebel	Nuclear Quantum Effects — How Much Does Chemistry Change?
OC22	16:15–16:30	Flávio Olimpio Sanches Neto	N-Nitrosation of a Sitagliptin-Related Secondary Amine by N ₂ O ₃ and NO ⁺ : A Benchmarked Mechanistic and Kinetic Study
OC23	16:30–16:45	Daniel Oliveira	Pressure-Dependent Kinetics of the NH ₃ + OH Reaction from Cryogenic to High Temperatures
OC24	16:45–17:00	Orlando Villegas	Systematic Search of Reaction Intermediates in Catalysis Using the GSAM Software and Machine Learning
OC25	17:00–17:15	José A. Quiñonero	AIM4ML: A Scaffold-Diverse Molecular Dataset for IQA-Informed Machine-Learned Force Fields
OC26	17:15–17:30	Ruben Van der Stichelen	Parameterized Landscapes in Quantum Computing Hartree–Fock: Implications for Benchmarking Noisy Quantum Devices
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Session I

Chair: Charlotte Titeca

Generalized Exchange and Correlation from Mayer's Improved Bond Order Formula

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The second-order reduced density (2-RDM) matrix is usually decomposed into Coulomb, exchange, and correlation terms, the latter originating from the cumulant. In the spinless formulation, the exchange contribution originates solely from the first-order density (1-RDM) and spin density matrices.

Historically, the bond orders (BO) have been tightly related to the exchange-correlation density, through its definition in terms of the covariances of the atomic populations.¹ The well-known Mayer BO formula, however, makes only use of the 1-RDM.² Many years after the introduction of his first BO equation, Mayer proposed an improved definition,³ replacing the spin density by an expression using the density of unpaired electrons of Takatsuka.⁴ The improved BO formula still avoids the use of higher-order density matrices, eliminates the dependency on the m_s value for non-singlet states, while exactly reverting to the original BO for restricted single-determinant wavefunctions.

The improved BO expression *de facto* represents a *spinless* alternative definition of the exchange part of the 2-RDM, thus providing a different separation of exchange and correlation. We refer to these alternative definitions as generalized exchange and correlation, respectively. Characterizing the differences between the classic and the newly proposed generalized exchange could have implications for many descriptors, such as bond orders, diatomic energy components within the interacting quantum atoms (IQA) framework,⁵ or electron correlation indicators.⁶

In this work, we present an assessment of the generalized exchange and correlation expressions for the aforementioned descriptors, considering several molecular systems in equilibrium and along their dissociation curves at the CASSCF level of theory. All formulae have been implemented in a newly developed Python version (pyAPOST) of the APOST-3D program.⁷

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What Does an Atomic Partition Need to be Reliable?

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The definition of the Atom-in-Molecule (AIM) is essential for understanding how atoms assemble into a molecule, identifying whether a bond is single, double, or triple, determining if electrons are more localized in specific regions, and even assessing whether a given molecular system exhibits aromaticity. Historically, real-space partitions such as the Quantum Theory of Atoms in Molecules (QTAIM)¹ have been commonly used because of their high degree of reliability, though they suffer from high computational costs and numerical integration errors in large systems.² On the other hand, so-called Hilbert-space partitions make use of atom-centered basis functions to describe the AIM, avoiding numerical integrations when using Gaussian-type orbitals.

Recently, we demonstrated that some Hilbert-space partition schemes, such as meta-Löwdin³, Natural Atomic Orbitals (NAO)⁴, and Intrinsic Atomic Orbitals (IAO)⁵, could serve as a computationally efficient, robust, and reliable alternative to real-space partition schemes for the calculation of electronic structure indicators.⁶

The IAO scheme has emerged as one of the most elegant and promising Hilbert-space partition, which generates a minimal basis that exactly spans the occupied space in single-determinant wavefunctions. However, this leads to a conundrum in the case of the π delocalization in cyclobutadiene: it is exactly zero for a minimal basis description, contrary to QTAIM. The harpoon effect poses another challenge for an AIM: during LiH dissociation, the system shifts from ionic bonding to neutral atoms, with electron sharing peaking at the crossing between the ionic and covalent states. Only those AIM schemes that gather information from the molecular electron density can reproduce this behavior.

In this work, we will explore different modifications of the IAO scheme to tackle the aforementioned limitations, namely i) the extension of the minimal basis to accommodate polarization effects, ii) the generalization of the IAO scheme to correlated wavefunctions, and iii) a density-dependent in-situ generation of the reference atomic basis. We will benchmark electronic structure indicators against QTAIM results, ensuring that the desired basis set independence and robustness are maintained.

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Atoms-in-Molecules Rényi Entanglement Entropies for Correlated Wavefunctions

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Electron entanglement in molecules provides an observable measure of correlation. Rényi generalizations of the accessible entropy S_{acc} - with the infamous von Neumann entropy as the special case $\alpha = 1$ -, quantify the information accessible for quantum information processing and quantum computation, and are in principle experimentally measurable.^{1,2} Rényi entanglement entropies are readily available from Quantum Monte Carlo calculations⁴ and provide complementary probes of the entanglement spectrum through a family of entropies parameterized by α . When an N -electron correlated wavefunction is partitioned in real space into atomic domains, particle-number superselection rules (N-SSR) block the reduced density matrix into sectors of fixed local electron count on each domain; S_{acc} - the probability-weighted sum of within-sector Rényi entropies - isolates the physically extractable quantum correlations from the inaccessible number-fluctuation contribution between domains. While work on this has been performed for single-determinant Hartree-Fock wavefunctions using solely the von Neumann entropy³, this measure was found to be independent of internuclear distance in H_2 , motivating the need to introduce correlated wavefunctions.

Thus, in this work we develop an exact framework for evaluating S_{acc} for correlated wavefunctions. We base our framework in the diagonalization of domain overlap matrices S^Ω and obtaining the resulting Domain Natural Orbitals (DNOs) with eigenvalues $d_k \in [0,1]$ that encode fractional electron localization. As a benchmark, we apply the framework to minimal-basis FCI and general-basis CAS(2,2) dissociation curves of two-electron toy models. We study the relations between Rényi entropies for different α and how they connect to the probability distribution of electrons in molecules and the intrinsic entropies of each sector. The Heitler-London limit of pure spin entanglement is correctly recovered at large internuclear distances, and the single-determinant limit of the entropies is also correctly described within our framework.

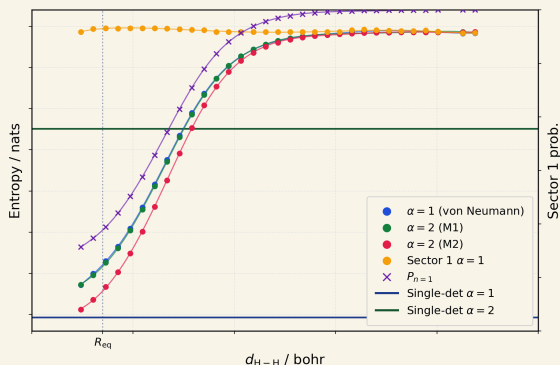


Figure 1. Accessible entropies (von Neumann and two versions¹ of Rényi $\alpha=2$, sector 1 entanglement entropy and sector 1 probabilities vs. H-H distance (FCI/STO-6G) for the dissociation of H_2 .

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Benchmarking TDDFT in Describing Excited State Bonding Analysis

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Over the past few years, chemical bonding in excited electronic states have emerged as a central topic of research, due to their relevance in analysing various photophysical and photochemical processes.^{1,2} Reliable theoretical methods to investigate the bonding nature of these states is essential. As we go to excited states (ESs), electron correlation becomes more prominent and defining a bond order becomes more complicated. Real-space bond order descriptors, delocalization indices (DIs) tackle this problem by using orbital-invariant techniques.³⁻⁵

While wavefunction-based methods like CCSD, CASSCF and FCI provide accurate description of ESs, they are computationally demanding and limited to smaller molecules.^{6,7} Time-dependent density functional theory (TDDFT) may provide an alternative at the cost of a lower accuracy. Some previous reports indicate that although DIs and other similar descriptors like I_{ring} , multicenter delocalization index have been applied with TDDFT computations but they lack proper benchmarking studies. The reliability of these values have not been assessed properly. We test the performance of TDDFT for a range of density functional approximations with respect to advanced wavefunction methods in addressing the DIs of excited states for a variety of systems.

We have studied different type of cases to address a range of excited state phenomena. We analysed the potential energy curves (PECs) of excited singlets and triplets of a range of diatomic molecules like H_2 , He_2 , LiH and LiF . DI values on CASSCF studies on excited singlet states give very distinctive results near the regions of avoided crossings between the covalent and ionic states. We are trying to see how TDDFT performs with respect to these wavefunction methods. We are looking into the DIs of the ground and the first excited states of ethylene along the dihedral rotation of the C-C bond. We are further investigating the extent of electron delocalization of a number of excited states for aromatic systems like benzene and biphenylene.

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Exotic Molecules Through the Lens of Quantum Chemical Topology: A Fresh Perspective on Chemical Bonding

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The design of molecular frameworks that mirror appealing objects has long been a wellspring of inspiration for chemists. Molecular knots, charge-shift-bonded species, and multicentric bonds, among many others, all display exotic properties rooted in an unconventional picture of chemical bonding, one that can even defy the classical Lewis model. Because these systems sit at the edge of standard bonding descriptions, they demand an interpretative framework able to recover chemically meaningful information directly from quantum mechanical observables.

Quantum Chemical Topology (QCT) offers exactly this. The analysis of scalar fields readily obtained from the wavefunction, namely the electron density, the electron localization function (ELF), or the Laplacian of the electron density, partitions the real space into chemically identifiable objects and quantifies the properties of the chemical bond. Complementary approaches such as the interacting quantum atoms (IQA) decomposition further dissect the system energy into physically meaningful contributions.

In this work, the QCT toolkit is applied to a selection of exotic systems. Among them, the pyramidane family, built on a $[C_4H_4]$ fragment and capped by an apical element that completes a pyramidal skeleton, has been investigated through the lens of ELF topology. The results reveal a pronounced electron delocalization captured by trisynaptic ELF basins. This phenomenon increases along the group 14 elements (C, Si, Ge, Sn and Pb) when they participate as the apex. Electron distribution functions (EDF) and IQA reinforce the qualitative picture through electron-counting statistics and interaction energies, respectively.¹ In the same spirit, the extremely electronegative *teflate* group, $[OTeF_5]^-$, has been studied, and its resemblance to fluorine chemistry has been unveiled: the IQA decomposition distinguishes genuine covalency from electrostatically driven bonding in its interaction with a bonded halogen atom.²

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Collective Interactions in Ion Pairs

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Herein, we report a quantum-chemical investigation into whether nominally ionic ion pairs can display collective bonding¹, a stabilization mode arising from distributed exchange-correlation interactions rather than a single two-center bond.

Using Penetration Indices² and Interacting Quantum Atoms (IQA) energy decomposition, we examined a series of ion pairs $M[AX_4]$ (Fig. 1; $M = Li^+ - Cs^+$; $A = B, Fe, Co, Zn, Cd$; $X = -CH_3, -CO, -CCH, -OCH_3, -Cl$). We found that alkali-metal tetramethylborates and several transition-metal tetrahedral anions show clear signatures of collective behaviour: the direct $M \cdots A$ interaction is weakly covalent or even repulsive, while stabilization is dominated by exchange-correlation contributions distributed over multiple $M \cdots X$ contacts.

Systems bearing polarizable or electron-rich ligands ($-CCH, -OCH_3, -Cl$) exhibit low interaction-collectivity indices (ICI_{XC} 0.07), it means, they exhibit collective behaviour, whereas carbonyl-containing complexes show substantially higher values (> 0.25), consistent with more localized bonding. These results extend the concept of collective bonding into ionic environments and suggest that ion pairing, solvation, and reactivity may often be governed by multicenter exchange-correlation effects rather than purely electrostatic or classical two-center interactions.³

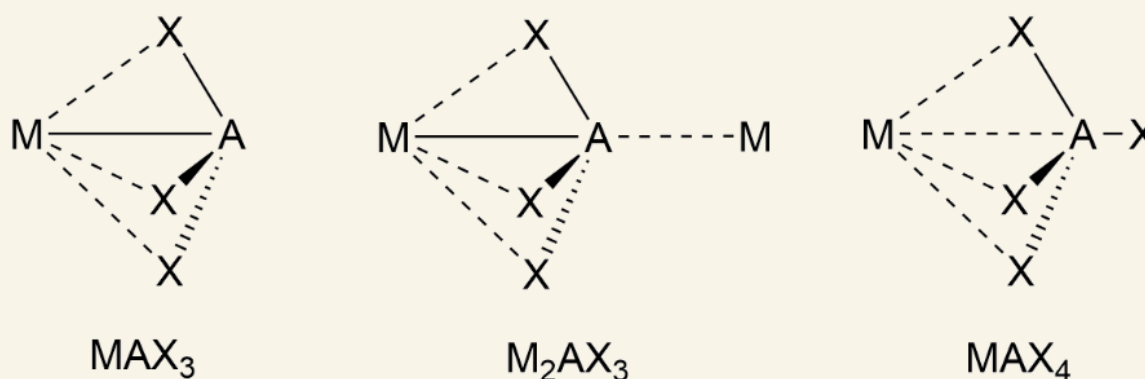


Figure 1. Structural motif of the studied ion pairs (MAX_4).

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Session II

Chair: Jessica Bowles

Looking Around the Dewar–Chatt–Duncanson Model: Environmental Effects in TM–CO Bonding

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For decades, the Dewar–Chatt–Duncanson (DCD)¹ model has provided the fundamental framework for understanding transition metal (TM)–ligand bonding, with metal carbonyls serving as a seminal example. The sensitivity of the carbonyl stretching frequency, $\nu(\text{CO})$, to its electronic environment has made it a widely used probe of ligand donor properties. However, recent computational advances have shown that $\nu(\text{CO})$ is influenced by factors beyond the electron-richness of the metal it is bound to, as exemplified by nonclassical carbonyl complexes.²

Recently, we discovered how electrostatic effects operating beyond the first coordination sphere exert a significant influence on TM–CO interactions in a series of late TM pybox carbonyl complexes.³ Building on these results, we have investigated the influence of local electric fields and non-covalent interactions on metal–carbonyl bonding using a combination of infrared spectroscopy, energy decomposition analysis (EDA), and ab initio molecular dynamics (AIMD).⁴ This combined experimental and computational approach allows us to disentangle environmental effects from the traditional orbital contributions to the TM–CO bond. These findings provide new design principles for ligand architectures and a broader perspective on the role of non-covalent interactions and local electric fields in TM–ligand bonding.

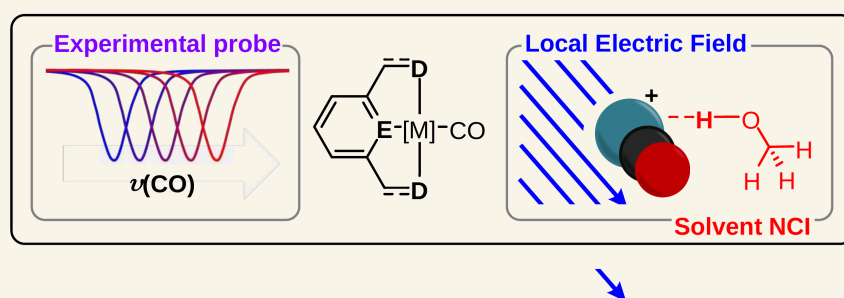


Figure 1. Overview of the concepts discussed.

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Molecular Structure and Bond Characterization of Fischer Boracarbene Complexes

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The unique electronic structure of transition metals (TMs), characterized by partially filled d orbitals, allows access to multiple oxidation states and coordination geometries, enabling complex processes beyond the reach of other elements.^{1,2} Fischer carbene complexes³ are versatile organometallic reagents in which the carbene carbon is bonded to a low-valent group VI-VIII TM through a formal metal-carbon double bond. This bond arises from synergetic σ donation of the carbene carbon lone pair to the $d(\sigma)$ orbital of the metal, and π back-donation from the metal's occupied $d(\pi)$ orbital to the carbene $p(\pi)$ orbital (Scheme 1, left), and is central to their reactivity.⁴

Concerns over the scarcity, cost, and environmental impact of TMs have spurred the development of main-group metallomimetics,⁵ in which main-group compounds behave analogous to TM complexes. These low-valent p-block species often exhibit TM-like orbital properties despite lacking the electronic flexibility of partially filled d orbitals, and undergoing only limited redox interconversion. Boron is particularly attractive in this regard: despite its non-metal character and low atomic weight, it is increasingly encountered in atypical oxidation states and bonding environments.² In this context, we have analysed the B=C and B=Si bonds in a series of 25 boracarbene and borasilylene complexes (Scheme 1, right). To quantify the degree of donation and back-donation in bora-Fischer carbene (B=C bond) and silylene (B=Si) complexes, we have combined Morokuma-like energy decomposition analyses (EDA) with wavefunction analysis tools present in the APOST-3D package.

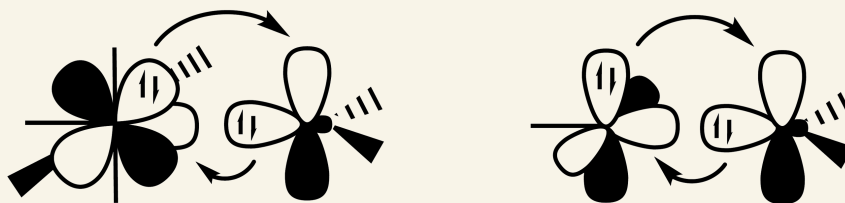


Figure 1. Representation of the metal-carbene bond of Fischer carbene complexes (left) and boracarbene complexes (right).

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Metallaaromatic Macrocycles via Dative Bonding: A Computational Study

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Metallaaromatic systems, in which transition-metal *d*-orbitals actively participate in π -conjugation, are typically constructed by replacing a carbon atom within an aromatic framework with a transition-metal center, as exemplified by metallabenzenes and metallapentalenes. Here, we present an alternative approach: formation of metallaaromatic macrocycles based on dative metal-ligand bonding, which allows for much more tunability and modularity in molecular construction.

We synthesized and experimentally characterized a new series of Rh, Fe, and Co pincer-ligated macrocycles, and computationally investigated their electronic structure, bonding, and aromatic properties. Using density functional theory (DFT), we show that dative metal-ligand coordination enables efficient electronic communication across the macrocyclic framework, incorporating the metal *d*-orbitals into the delocalized π -system to form electronically coherent conjugated circuits. By comparing different metal centers and coordination environments, we explore how the nature of the metal-ligand bonding interaction governs the degree of macrocyclic delocalization and metallaaromatic character. NICS, ISE, and magnetically induced current density analyses reveal sustained global diatropic ring currents, consistent with robust metallaaromaticity.

These results establish dative-bonded macrocycles as a new platform for metallaaromaticity beyond traditional atom-replacement approaches, while opening a broader chemical space for investigating the interplay between chemical bonding, transition-metal electronic structure, and aromatic delocalization.

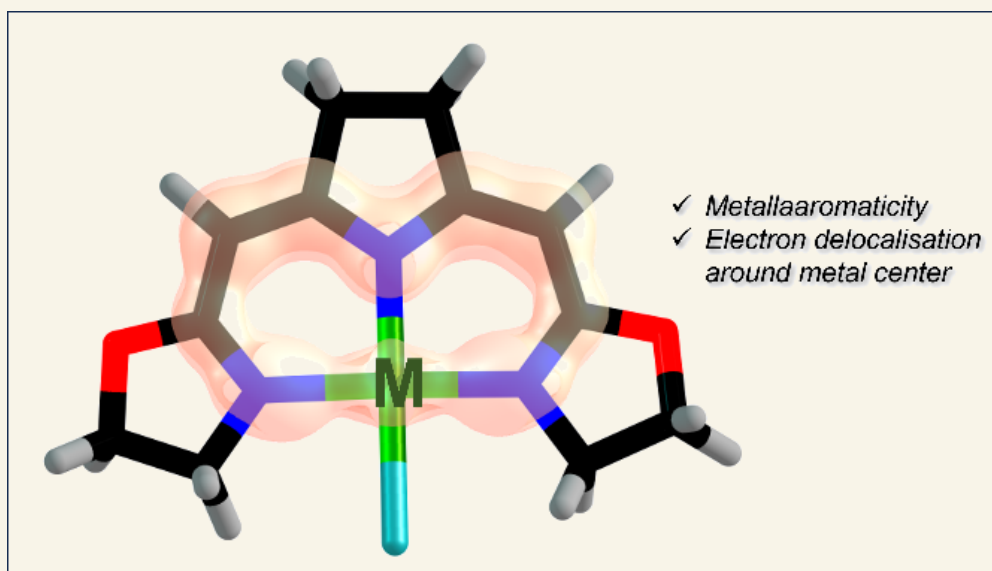


Figure 1. Dative bonding induced electron delocalization around metal center

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Beyond the Aromaticity–Diradical–Conductance Triangle

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We have investigated the fundamental relationships within the diradical character–aromaticity–conductance triangle (Figure 1) across oligothiophenes capable of adopting aromatic and quinoidal structures.¹ A strong correlation between the multicenter index of the central thiophene ring(s) and the diradical character (y_0) highlights the intrinsic connection between the aromaticity and diradical character vertices. The diradical character–conductance relationship reveals two competing effects: while higher generally enhances conductance, this trend is counterbalanced by length-dependent attenuation, leading to near-zero conductance in sufficiently long oligomers (~ 20 Å).² Systems with exhibit reduced conductance regardless of molecular length. Conductance also correlates with local aromaticity: annulated rings that disrupt the aromaticity of the backbone thiophene units yield higher conductance compared to parent structures. Approximately 70% of local transmission pathways align with aromaticity descriptors, underscoring their predictive value for local charge transport behavior.³ Our study extends this triangle by incorporating the electronic ground-state structure as a fourth vertex and distinguishing between local and global aromaticity–conductance relationships, offering deeper insight into how electronic structure governs charge transport.

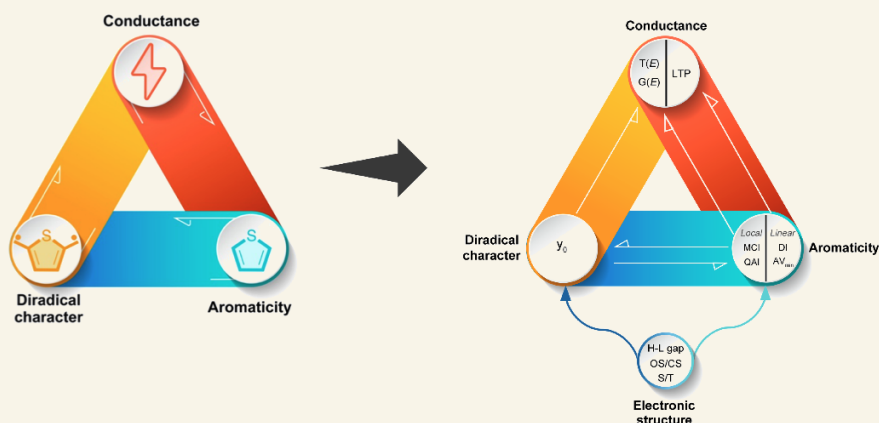


Figure 1. Schematic overview of the interplay between diradical character, aromaticity, and conductance, as well as the extension of set triangle.

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Computational Elucidation of Ligand Driven Selectivity in Gold(I) Catalyzed Alkyne Hydroamination

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A theoretical investigation was carried out to rationalize the reactivity and selectivity of L-shaped and V-shaped gold(I) catalysts based on anthracene (Anth) and endoperoxide (Endo) ligands. The modulation of the catalyst structure by singlet oxygen (¹O₂) provides an original platform to probe structure–reactivity relationships through quantum chemical modeling.

Density Functional Theory (DFT) calculations were used to examine the well described alkyne hydroamination mechanism¹ (Figure 1), with a focus on the initial nucleophilic attack step and the rate limiting step of protodeauration. Intermediates and transition states were fully optimized for both catalysts, considering the competing attack pathways of an aniline on both carbon atoms of the activated alkyne. Comparative analysis of the resulting energy profiles reveals a strong dependence of activation barriers and regioselectivity on ligand topology.

To gain deeper insight into the origin of these differences, QTAIM, NCI, and NBO analyses further highlight how ligand-induced steric and electronic effects modulate substrate activation and transition-state stabilization.

This presentation discusses the computational approach and mechanistic insights underlying reactivity differences between Anth- and Endo-based gold(I) catalysts.

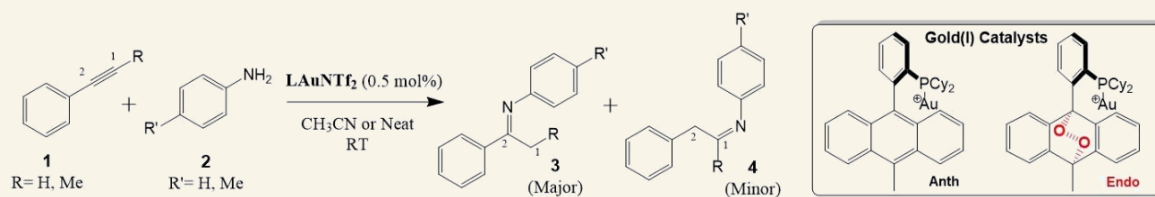


Figure 1. Hydroamination of alkynes **1** (R=H or Me) by aniline derivatives **2** (R'= H or Me) catalyzed by L-shaped gold(I) **Anth** or V-shaped gold(I) **Endo** catalysts.

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A Dynamic Evaluation of Cisplatin Encapsulation into $[\text{Pd}_2\text{L}_4]^{4+}$ Metallacages in Solution

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Coordination metallacages (MCgs) enable the design of various host structures capable of binding therapeutic guest molecules, such as cisplatin. In this study, well-tempered metadynamics (WT-MetaD) simulations¹ were performed in explicit water and DMSO to investigate the host–guest interactions between two homoleptic $[\text{Pd}_2\text{L}_4]^{4+}$ cage^{2,3} (*endo*-C and *endo*-N), differing in the hydrophilic character of their cavities, and cisplatin. The free-energy landscape revealed that the most stable state of the MCg⊃cisplatin systems corresponds to two cisplatin molecules associated with the external MCg surface (state I), via van der Waals or H-bonding interactions, regardless of the *endo*-functionalization. The different NCI established in the different states were analysed in detail via NBO, NCI index and QTAIM. The analysis of observed MCg⊃cisplatin adducts suggests a transient association of the drug with the cage scaffold rather than tight encapsulation within the cavity.

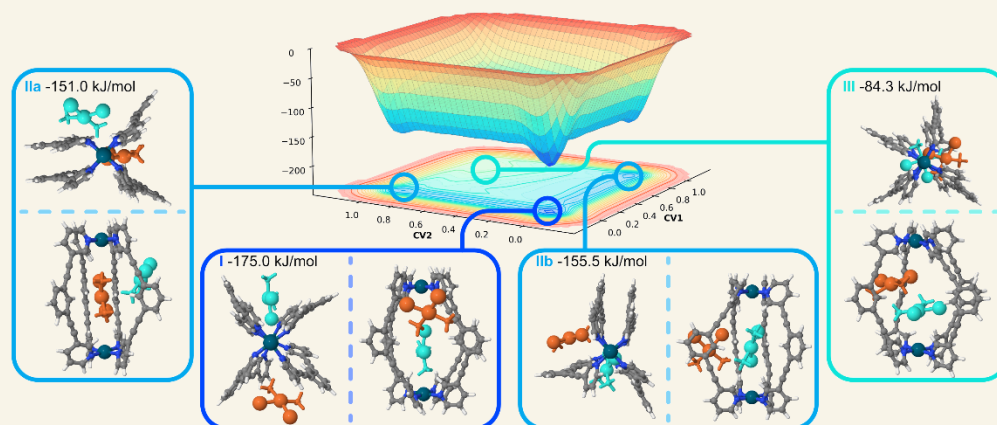


Figure 1.

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Session III

Chair: Irene Casademont-Reig

From Linear to Nonlinear Stark Responses: An IQA Perspective on Carbonyls and Nitriles

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The stretching frequencies of carbonyl (C=O) and nitrile (CN) groups are widely used as infrared probes of local electrostatic environments due to their sensitivity to electric fields.¹ Carbonyls typically exhibit a consistent red shift in response to their surroundings, in line with the linear vibrational Stark effect.² In contrast, nitriles display a more complex, non-linear behavior, including blue shifts in protic solvents, making them spectrally richer but less predictable.^{2,3} Although energy decomposition schemes, Valence-Bond Theory, and other methods have revealed competing interactions, a direct readout identifying the origin of the energy response is still lacking. To this end, we leverage the QTAIM/IQA^{4,5} framework to obtain an interaction-resolved account of how solvation reshapes the vibrational PES along the C=O/CN stretch coordinate in microsolvated acetophenone, benzonitrile, and methyl thiocyanate. Our results suggest that for all probes, irrespective of the solvent type, the on-bond exchange–correlation interaction remains constant, and so the response emerges not from changes in covalent bonding strength but from the competition between a few classical interactions. Carbonyls' behavior depends mostly on one deformation interaction. This single-interaction dependency gives rise to the conventional linear Stark behavior of these bonds. In contrast, nitriles depend on Coulomb and framework contributions competing and canceling each other out, which results in both blue and red shifts. This means that the different probe responses arise from the same mechanism but with distinct response dimensionality. Specifically, bond order sets the response amplitude, while donor geometry selects its sign.

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Contrasting Effects of Triazole Substituents Used in Hydrogen, Halogen and Chalcogen Bonding

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Non-covalent interactions, such as hydrogen, halogen and chalcogen bonding have garnered recent attention for their applications in anion binding. Due in part to their simple synthesis, triazole-derived hydrogen, halogen and chalcogen bond donors are often used as anion receptors. Modulation of the electronic properties of the triazole backbones can be readily achieved through addition of different *N*-substituents. Mancheño has previously investigated the effect of the *N*-substituent on the strength of chloride anion binding for a series of hydrogen bond donor bis-triazoles.¹ It was found that addition of the electron-withdrawing *N*-3,5-(CF₃)₂-C₆H₃ group led to an increase in anion binding strength over *N*-Ph, whilst *N*-C₆F₅ gave the lowest binding strength of the three. This observed difference was believed to be due to the presence or absence of additional secondary interactions between the chloride and the *ortho*-protons of the *N*-substituent. Beer observed that for halogen and chalcogen bond donor triazoles that the *N*-C₆F₅ substituent showed a significant increase in anion binding strength over *N*-Ph, however an explanation for this difference was never explored.² The contrasting effects of these different substituents leads to the question of how these non-covalent bonding types differ intrinsically. This work aims to explore the role of the *N*-substituent, in monodentate triazole systems, on the anion binding strength of hydrogen, halogen and chalcogen bonding interactions, using both computational and experimental methods. Through NMR titrations a new 'best-in-class' *N*-substituent has been found for halogen and chalcogen bond donor triazole systems. The non-covalent interaction (NCI) method coupled with quantum theory of atoms in molecules (QTAIM) analysis has uncovered the strengths of the primary and secondary interactions in this series of compounds.^{3,4} Computational analysis has also revealed that modulating the tellurium and triazole substituent of chalcogen containing triazoles leads to different anion binding angles which could be exploited experimentally in the future. This analysis has helped identify key design features which can be varied to tune future generations of hydrogen, halogen and chalcogen bond donor anion receptors and catalysts.

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DFT Insights into Noncovalent Stabilization of Cationic Intermediates in Pillar[5]arene-Catalyzed Anti-Markovnikov Halogenation

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The confinement of reactive intermediates within well-defined molecular cavities offers a powerful strategy for controlling reaction pathways and overriding intrinsic selectivity. Here, pillar[5]arene-catalyzed anti-Markovnikov halogenation of olefins is investigated through density functional theory calculations to elucidate how noncovalent interactions within a π -basic confined cavity stabilize cationic bromiranium intermediates and direct regioselective nucleophilic attack.¹ The calculations show that the olefin, brominating reagent, carboxylic acid, and pillar[5]arene host assemble into a reactive complex in which bromonium transfer to the encapsulated olefin occurs with a low activation barrier, generating a confined bromiranium intermediate. Subsequent structural reorganization within the cavity positions the cationic intermediate for preferential nucleophilic ring opening at the less substituted carbon, thereby favoring the anti-Markovnikov product.

The computed energy profile indicates that both bromiranium formation and nucleophilic ring opening are kinetically accessible, supporting the catalytic role of pillar[5]arene in stabilizing the cationic intermediate rather than merely binding the substrate. Noncovalent stabilization from the π -basic cavity, together with steric shielding of the more substituted carbon, provides a mechanistic basis for the reversal of the conventional Markovnikov preference. These DFT results rationalize the experimentally observed regioselectivity and size selectivity, demonstrating that confined-space stabilization of cationic intermediates can be exploited as a general design principle for selective electrophilic transformations.

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Theoretical Study of Monohydrated Molecules Bearing Sulfur and Oxygen Atoms

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In the second half of the 20th century, several volatile organosulfides were discovered in the atmosphere. Among them, dimethylsulfide (DMS) was identified as the primary molecule responsible for the sulfur cycle¹. Characterizing the microhydration of different sulfur-containing molecules may help explain specific atmospheric phenomena². In the present work, we were focused on the monohydration of the species.

Quantum calculations were performed for monohydrated key molecules in the sulfur cycle—DMS, DMSO, OCS, H₂S, H₂CS and CS₂—using LC- ω PBE/aug-cc-pVDZ. The same computational methods were applied to their oxygen analogues—H₂CO, (CH₃)₂CO, (CH₃)₂O, CO₂, and H₂O—to provide a basis for comparison.

At equilibrium, a Molecular Electrostatic Potential (MEP) study was conducted to identify nucleophilic and electrophilic sites. After optimizing the monohydrated complexes using LC- ω PBE/aug-cc-pVDZ, a Quantum Theory of Atoms in Molecules (QTAIM)³ analysis was performed to locate intermolecular bond critical points (BCPs) and to characterize the bond strength. A SAPT (Symmetry Adapted Perturbation Theory) analysis was finally carried out to gain some insight in the origin of the stability of the complexes.

The results indicate that S...H hydrogen bonds in monohydrated complexes are weaker than their O...H counterparts. Moreover, a negative linear correlation was observed between the complexation energy and the electron density at the BCP of the intermolecular hydrogen bond. SAPT analysis shows that oxygen-containing complexes are dominated by electrostatic interactions, whereas sulfur-containing systems exhibit a much larger dispersion contribution, sometimes becoming dominant. This difference arises from the higher polarizability of sulfur, which enhances dispersion interactions.

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Combining Experimental and Theoretical Raman Spectroscopy to Reveal Topology and Bonding in Soluble Silica

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Silica's speciation has proved to be a complex phenomenon with far-reaching implications. Techniques capable of characterizing silica oligomers are paramount due to the diversity of applications involved, from functional additives to the non-classical nucleation routes of silica-based materials. ²⁹Si NMR remains the most successful technique for the characterization of silica's speciation. However, its limited applicability beyond research underscores the need for a powerful, sustainable and versatile alternative with in situ characterization potential.¹ Raman spectroscopy has proven its potential to fulfill these requirements.

In this work, we tackled a decades-old challenge: the unique assignment of Raman features in silica oligomers. Integrating experimental Raman measurements with state-of-the-art ab initio dynamic Raman simulations and conformationally averaged static DFT calculations, we deconvolute the complex vibrational response of silica-containing solutions. Our approach explicitly incorporates largely overlooked theoretical contributions, such as anharmonicity, realistic solvation, and conformational diversity, enabling robust assignments of Raman bands to individual species.² Moreover, using the same integrated approach, we complete the database of Raman signals for most of silica oligomers, allowing for the discovery and rationalization of the underlying trends in their spectra, guided by their bonding and topological nature.³ Furthermore, we address the effect of protonation in the spectra, and the common use of inappropriate notations to describe the vibrations of such oligomers.

Bringing Raman spectroscopy into the toolbox for the determination of silica's speciation offers two key advances. First, real-time in situ characterization can significantly deepen our understanding of nucleation and growth pathways in silica-based systems. Second, improved control over synthesis enables the development of more sustainable synthetic protocols, by reducing waste associated with unwanted by-products, while accelerating the discovery of novel materials with promising functional properties.

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Defect Mediated Surface Modification of TiO₂ Surface Using Grignard Reagent in Explicit Solvation Model

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Titanium dioxide (TiO₂) is widely used in catalysis and photocatalysis due to its versatile surface chemistry¹. Among its crystalline polymorphs, anatase TiO₂ exhibits superior catalytic performance, largely attributed to its high surface area and undercoordinated active sites. In addition, amorphous form of TiO₂ plays a crucial role in sorption, separation, and catalytic technologies owing to its structural disorder, enhanced surface area, and electronic mobility. Conventional surface functionalization relies on oxygen-based anchoring groups², organometallic approaches such as Grignard reagents offer a more direct route to covalent modification through Ti-C or Ti-O-C bond formation.³⁻⁴

Here, we use DFT+U calculations to study the adsorption and reactivity of methyl magnesium bromide (CH₃MgBr) on anatase and amorphous TiO₂ surfaces, including pristine and defected models with oxygen⁵, titanium, and combined Ti-O vacancies. We further investigate the effect of explicit THF solvent using ab initio molecular dynamics. Our results show that both surface defects and solvent coordination strongly influence adsorption behavior and stabilize reactive intermediates, facilitating Ti-C bond formation. These findings provide mechanistic insight into solvent- and defect-assisted Grignard functionalization of TiO₂ and offer guidelines for targeted organometallic surface engineering.

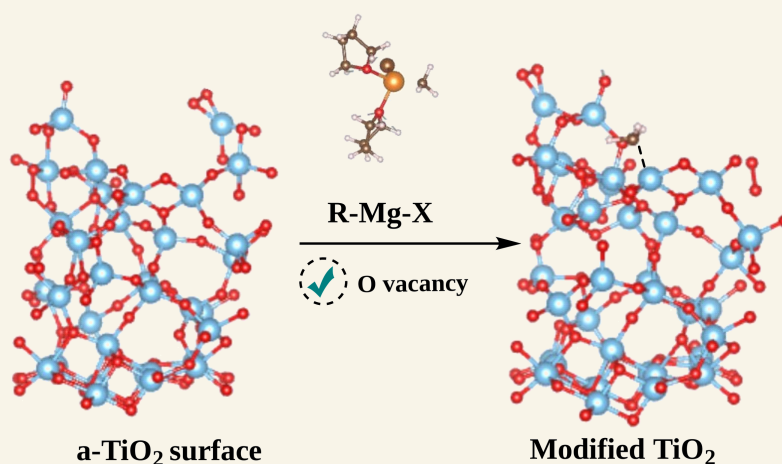


Figure 1. Surface modification process of amorphous titania(a-TiO₂) using Grignard reagent(RMgX)

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Chemical Bonding Information from High Pressure Chemistry

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High pressure chemistry has come a long way since its initial conception, e.g. under the form of ball-milling. Its use in organic and inorganic synthesis has been well documented¹ and more recently plenty of new materials, routes towards them and complicated electronic structures have been discovered by making use of high pressure. While methods to simulate high pressure chemistry such as periodic DFT and MD simulations have existed for a while, cheaper alternatives for static calculations such as XPPCM, XHCFF and GOSTSHYP have been introduced only relatively recently. Using these models, chemical concepts can be studied on archetypal systems from a general point of view, resulting in chemical information from the energy domain that regardless has applications to chemical bonding. When focusing on atoms under an isotropic pressure, electronic volumes, trends in electronegativity and hardness and rules on electrone formation can be straightforwardly identified.² Transitioning to diatomic molecules, bond surfaces can be obtained from the pressure response of equilibrium bond lengths.³ These properties are also accessible analytically through a linear response formalism, as is shown for dipole moment derivatives under pressure. Finally, the influence of pressure on aromaticity is evaluated for the archetypal aromatic compound: benzene.⁴ During the evaluation of these energy derived properties, particular attention is dedicated on the pitfalls when trying to determine them.

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Session IV

Chair: Bin Wang

Conceptual DFT for and from the Excited States

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The Conceptual DFT (CDFT) is broadly interpreted as a branch of the DFT to make link between quantum calculation and chemical properties. Nevertheless, CDFT is based on the Hohenberg and Kohn theorems meaning that CDFT reactivity descriptors can only be used to describe the ground state. Since chemistry is not just about the ground state — photochemistry, for example, deals with excited states — extended the CDFT for these states would be of a great interest.

Descriptors of CDFT are expressed as response functions to perturbations either in the external potential or the number of electrons. Following the idea of a perturbation of the external potential, in 2020 Guégan et al. used the Rayleigh-Schrödinger perturbation theory (RSPT) coupled with TDDFT to formulate a set of polarization descriptors.^{1,2}

The interest in this approach is that RSPT is formulated in any state, hence the polarization descriptions can be quite straightforwardly extended to excited states. In this presentation, I will illustrate how such an extension can be provided, and show the potential of the development on some prototypical molecular systems. I will also show how an alternative development, relying on the relation between the polarization descriptors and the linear response function of CDFT,³ can help in addressing the issues RSPT sometimes present (deviation from weak perturbation regime).

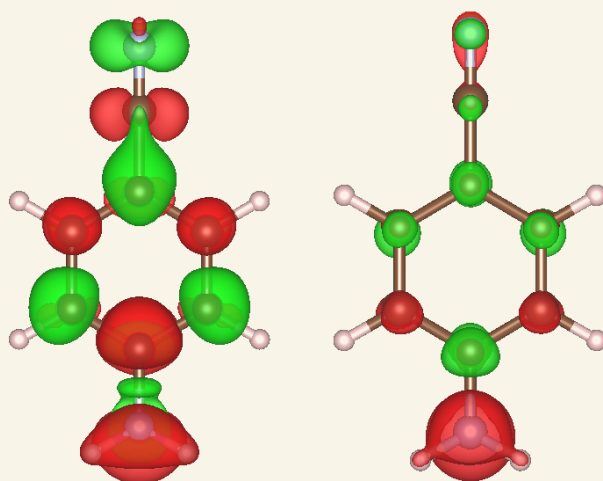


Figure 1. Polarization response of the electronic density of para-cyanoaniline in the ground state (left) and the charge transfer excited state (right).

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Nuclear Quantum Effects — How Much Does Chemistry Change?

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Treating nuclei as classical point charges has underpinned many advances in computational chemistry. Yet for hydrogen nuclei, this approximation introduces a notable blind spot: Nuclear Quantum Effects (NQEs) govern proton transfer and proton-coupled electron transfer, and reshape the geometry and strength of hydrogen bonds; both are central to processes such as enzyme catalysis.^{1,2}

Multicomponent methods address this limitation. Selected nuclei, most commonly protons, are treated as quantum particles on equal footing with electrons. The Nuclear-Electronic Orbital (NEO) framework³ is one such approach. When applied to density functional theory (NEO-DFT)⁴, capturing effects such as proton delocalization and anharmonic zero-point vibrational energy becomes viable even for larger systems.⁵

Here we present a benchmark dataset of multicomponent DFT protonic densities and energies comprising 28 molecular systems, generated from thousands of DFT single point calculations per system, and validated against CCSD(T) results. We examine systematically how chemical environment governs the degree to which protons behave as classical point charges. This behavior is strongly shaped by the surrounding electron density and can disrupt established scaling relations across the spectrum from acidic to hydride systems.⁶ Selected examples illustrate how a quantum mechanical description of protons affects chemical properties and associated descriptors.

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N-Nitrosation of a Sitagliptin-Related Secondary Amine by N_2O_3 and NO^+ : A Benchmarked Mechanistic and Kinetic Study

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N-Nitrosamines are genotoxic impurities of major regulatory concern in pharmaceuticals, and the nitrosamine drug substance-related impurities (NDSRIs) derived from secondary-amine active ingredients are especially difficult to assess because compound-specific data are scarce.¹ The triazolopyrazine TTP, a degradant of the antidiabetic sitagliptin, is converted to the genotoxic impurity NTTP under conditions relevant to manufacture and storage.² While the N-nitrosation of secondary amines by N_2O_3 has been examined computationally,³ a systematic, high-level mechanistic and kinetic description for TTP is still lacking. Here we characterize the N-nitrosation of TTP by the three N_2O_3 isomers (asym, sym, trans-cis) and by NO^+ , combining electronic-structure benchmarking with transition-state kinetics.

Conformers of TTP and of the N_2O_3 isomers are generated by conformational sampling. Reactants, pre-reactive complexes, transition states (TS) and products are optimized with a set of density functionals (B3LYP-D4, M06-L, M06-2X, ω B97X-D4) and basis sets in implicit solvent (SMD, water), complemented by explicit microsolvation where proton transfer is involved. Each TS is confirmed by a single imaginary mode and by intrinsic-reaction-coordinate analysis, and Gibbs free energies are obtained at a 1 mol L⁻¹ standard state with a quasi-RRHO treatment of low-frequency modes. The functionals are benchmarked against DLPNO-CCSD(T) energies extrapolated toward the complete basis set, and rate constants follow from transition-state theory with deformed-TST tunneling.

The N-nitrosation is found to proceed through a concerted, base-assisted mechanism: a five-membered-ring TS in which the new N–N bond to the amine forms as the amine N–H proton is transferred and the N–N (or N–O) bond of the nitrosating agent breaks, releasing nitrous acid (N_2O_3 route) or protonated water (NO^+ route). The three N_2O_3 isomers and the NO^+ pathway are compared on a common footing, identifying the kinetically preferred agent and ranking the competing routes by their computed rate constants in solution.

These results provide a validated electronic-structure and kinetic basis for NTTP formation. As a perspective, the framework will be coupled to the CYP2E1-mediated α -carbon hydroxylation of NTTP (QM/MM) and the diazonium reactivity,⁴ and confronted with experimental quantification and mutagenicity data, toward a transferable scheme for NDSRI risk assessment.

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Pressure-Dependent Kinetics of the $\text{NH}_3 + \text{OH}$ Reaction from Cryogenic to High Temperatures

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The $\text{NH}_3 + \text{OH}$ reaction, which forms NH_2 and H_2O , is of great interest in the interstellar medium, yet a comprehensive theoretical description for cryogenic temperatures and wide pressure ranges is lacking. To address this, we combined electronic structure calculations (CCSD(T)/CBS//MPWB1K/6-31+G(d,p)) with chemical kinetics. Rate calculations utilized the canonical unified statistical (CUS) theory (including CVT/SCT), while pressure effects between 20–2500 K were treated using system-specific quantum RRK (SS-QRRK) theory.¹ The results reveal a typical hydrogen abstraction mechanism involving weak pre- and post-reactive complexes. High-pressure rate constants show good agreement with experimental data² over a wide temperature range, although deviations at very low temperatures highlight the importance of pressure effects. Pressure-dependent calculations reproduce experimental values in the 0.1 to 0.01 mbar range, more specifically, at 22 K, the experimental value is 8.4 times higher than our result ($2.3 \times 10^{-10} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$) and only 1.1 times higher than the estimate at 0.01 mbar ($3.0 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$), and show significant falloff behavior at low pressures due to inefficient collisional stabilization. Overall, this study provides a consistent description of the temperature- and pressure-dependent kinetics of the $\text{NH}_3 + \text{OH}$ reaction, offering reliable predictions for both atmospheric and interstellar conditions.

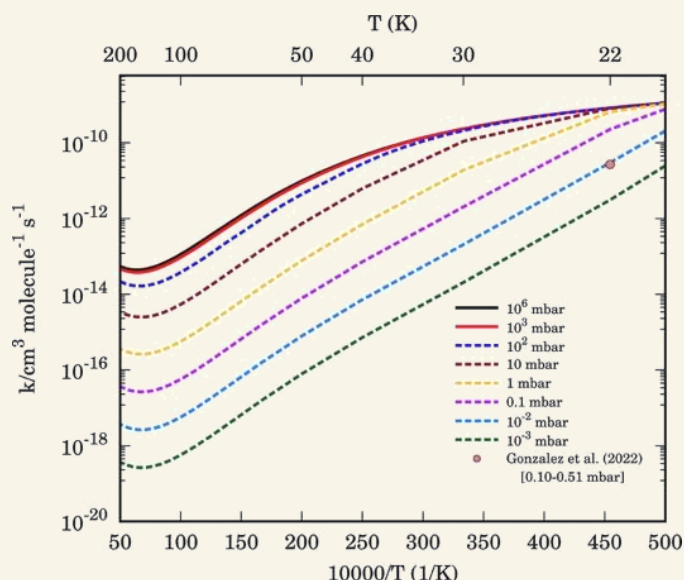


Figure 1. Low-temperature rate constants at 10^6 , 10^3 , 10^2 , 10, 1, 0.1, 10^{-2} , and 10^{-3} mbar, considering N_2 as the bath gas.

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Systematic Search of Reaction Intermediates in Catalysis Using the GSAM Software and Machine Learning

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Identifying reaction intermediates in catalysis requires the systematic exploration of high-dimensional potential energy surfaces. Conventional approaches based on molecular dynamics or random sampling are insufficient to cover the full conformational space of systems with diverse bonding topologies. Genetic algorithms (GA) offer an effective alternative for global structure search¹, but their coupling with first-principles methods entails prohibitive computational cost. We present GSAM (Genetic Search Algorithm of Minima Exploration), a software for the systematic exploration of reaction intermediates in catalysis. GSAM implements a GA that evolves populations of structures using molecule-specific mutation operators (motif decomposition, rotation, translation, and atomic exchange). Candidate structures are filtered by successive graph-theory and Coulomb matrix descriptors², which efficiently discard geometric duplicates and energetically unfeasible structures. To reduce the cost of local optimization, GSAM integrates an active learning (AL) cycle based on the MACE machine learning interatomic potential³. In each AL cycle, a representative subset of structures is optimized at the DFT level, the data is added to the training set, and the MACE model is refined via transfer learning until convergence in energy and force predictions is achieved. Applied to like $\text{FeO}^+ + \text{CH}_4$ in gas and others examples (Fig 1), GSAM successfully identifies previously reported intermediates alongside novel low-energy structures, while the MLIP active learning cycle achieves convergence with significantly fewer DFT calculations without sacrificing accuracy. The combination of evolutionary algorithms, geometry-based filters, and active learning with MACE enables efficient exploration of conformational space at an affordable computational cost, with direct applications in catalyst design and reactive surface studies.

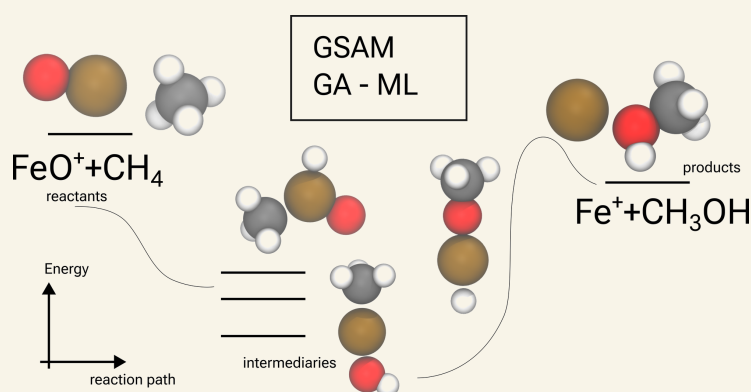


Figure 1: Search for intermediates in the reaction between FeO^+ and CH_4 .

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AIM4ML: A Scaffold-Diverse Molecular Dataset for IQA-Informed Machine-Learned Force Fields

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The Interacting Quantum Atoms (IQA) framework provides a rigorous real-space energy decomposition into intra-atomic and pairwise interatomic contributions,¹⁻³ yet no scaffold-diverse dataset carrying IQA annotations exists for training machine learning models at general-purpose scale.⁴⁻⁵

We present a generalizable, modular pipeline for curating and selecting molecular datasets for IQA annotation. A nine-stage curation module — performing SMILES canonicalisation, bond-order assignment, enantiomer filtering, and integrity validation — has been validated on QM40⁶ and is being applied to additional source databases including OMol25 to build a large, chemically diverse candidate pool. A diversity-based selection module (scaffold grouping + MaxMin sampling) will extract a representative subset balanced across chemical space. IQA exchange-correlation energies, atomic multipoles, and QTAIM topological indices will be computed with PROMOLDEN for the selected pool, with initial PM7 calculations already underway.

The resulting AIM4ML dataset will enable training of equivariant neural network potentials with physically interpretable, atom-wise energy components bridging QCT and modern ML architectures.

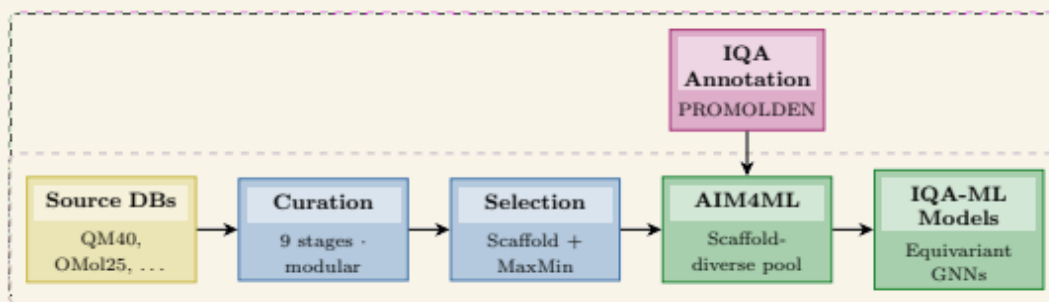


Figure 1. Modular pipeline for curating and selecting IQA-enriched molecular datasets.

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Parameterized Landscapes in Quantum Computing Hartree–Fock: Implications for Benchmarking Noisy Quantum Devices

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Recent experiments have demonstrated that quantum circuits can reproduce classically obtained Hartree–Fock (HF) solutions with the aid of error mitigation, where the focus lies on recovering a single optimized state from a noisy quantum device.¹ From a quantum chemical perspective, however, HF is defined as an energy minimization over a continuous orbital rotation space, whose convergence depends on the structure of the full orbital rotation landscape.² As such, validating a quantum implementation of Hartree–Fock requires probing more than a single optimized solution.

In this work, we examine how the parameter landscape is represented on noisy quantum devices. Using a parameterized single-determinant quantum circuit, we explore the orbital rotation manifold and characterize noise-induced deviations in energy and spin observables across the parameter landscape. We find that these errors are strongly non-uniform, even when mitigation strategies are applied, demonstrating that accuracy at an optimized point is not representative of the full variational landscape.³

In this talk, I use the parameter landscape of the quantum computing implementation of Hartree–Fock to show that benchmarking must extend beyond a single solution to accurately assess overall performance on noisy quantum devices.

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